

1. Consider the following codes:

```
public class MovieTicketCounter {  
  
    private int ticketPrice = 300;  
    private int popcornPrice = 120;  
    private int drinkPrice = 50;  
  
    MovieTicketCounter() {  
        System.out.println("Welcome to Movie  
Ticket Counter");  
    }  
  
    void purchase(int tickets, int drinks) {  
        System.out.println("You purchased " +  
            tickets +  
            " tickets, " +  
            drinks +  
            " drinks");  
        int total = tickets * ticketPrice +  
            drinks * drinkPrice;  
        System.out.println("Total bill: " +  
            total +  
            " taka");  
    }  
  
    // some more code  
}
```

```
class Customer {  
  
    public static void main(String[] args) {  
  
        MovieTicketCounter m = new  
MovieTicketCounter();  
  
        m.purchase(2, 4);  
        m.purchase(1, 2, 1);  
        m.purchase(3);  
    }  
}
```

Write the necessary missing codes so that the following output is produced when the program runs:

```
Welcome to Movie Ticket Counter  
You purchased 2 tickets, 4 drinks  
Total bill: 800 taka  
You purchased 1 ticket, 2 drinks, 1 popcorn  
Total bill: 520 taka  
You purchased 3 tickets  
Total bill: 900 taka
```

2. What is probably wrong with the following code? Explain briefly.

```
package utilities;

public class Calculator {

    int add(int a, int b) {
        return a + b;
    }
}

package application;

import utilities.*;

public class Main {

    public static void main(String[] args) {

        Calculator calc = new Calculator();
        // trying to access non-public method
        calc.add(10, 20); // not using the value
    }
}
```

3. Find out if the following Java programs have any errors. Fix the code and rewrite it. You cannot delete any line of code. However, you are allowed to edit or add code as required.

```
public class Employee {

    static int companyBonus = 5000;
    // int salary = 30000;
    // static method trying to access non
    // static attribute, hence make it static
    static int salary = 30000;
    // private int @id = 101;
    // invalid name
    private int id = 101;
    private String department = "HR";

    public static void showDetails() {

        System.out.println("Total: " +
            (companyBonus + salary));
    }

    // private attribute needs getter to be
    // accessed in other class
    String getDepartment() {
        return this.department;
    }
}

public class Office {

    public static void main(String[] args) {

        Employee e = new Employee();

        Employee.showDetails();

        // System.out.println(e.department);
        // private, access with getter
        System.out.println(e.getDepartment());
    }
}
```

4. Carefully consider the following program and write its output.

```
class DigitalWallet {

    private String owner;
    private double balance;

    DigitalWallet(String ownerName, double openingBalance) {
        owner = ownerName;
        balance = openingBalance;
    }

    void deposit(double amount) {
        balance = balance + amount;
    }

    void withdraw(double amount) {
        if (amount <= balance) {
            balance = balance - amount;
        }
    }

    double getBalance() {
        return balance;
    }

    String getOwner() {
        return owner;
    }
}

class Test {

    public static void main(String[] args) {

        DigitalWallet johnWallet = new DigitalWallet("John Smith", 1000.0);
        // balance = 1000.0
        DigitalWallet emmaWallet = new DigitalWallet("Emma Wilson", 200.0);
        // balance = 200.0
        johnWallet.withdraw(350.0);
        // balance = 650.0
        emmaWallet.deposit(500.0);
        // balance = 700.0

        System.out.println( johnWallet.getOwner() + " " + johnWallet.getBalance());
        // John Smith 650.0 ----- Output line 1
        System.out.println( emmaWallet.getOwner() + " " + emmaWallet.getBalance());
        // Emma Wilson 700.0 ----- Output line 2
        emmaWallet = null; // empties the reference but does not affect output
    }
}
```

```
    }  
}
```

5. Write the output of the given code.

```
public class Employee {  
  
    static int bonus = 1000;  
    int salary = 20000;  
  
}
```

```
public class Office {  
  
    public static void main(String[] args) {  
  
        Employee emp1 = new Employee();  
        // emp1.bonus = 1000, emp1.salary = 20000  
        Employee emp2 = new Employee();  
        // emp2.bonus = 1000, emp2.salary = 20000  
  
        emp1.bonus = 1500;  
        // emp1.bonus = 1500, emp1.salary = 20000  
        // emp2.bonus = 1500, emp2.salary = 20000  
  
        emp1.salary = 25000;  
        // emp1.bonus = 1500, emp1.salary = 25000  
  
        System.out.println(emp2.bonus); // 1500  
        System.out.println(emp2.salary); // 20000  
        System.out.println("Amount: " + 5 + 10);  
        // Amount: 510  
        System.out.println('B' + 3);  
        // 69  
    }  
}
```